

The red flower wintersweet genome provides insights into the evolution of magnoliids and the molecular mechanism for tepal color development

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SUMMARY

Wintersweet (*Chimonanthus praecox*) is one of the most important ornamental plants. Its color is mainly determined by the middle tepals. However, the molecular mechanisms underlying the intriguing flower color development among different wintersweet groups are still largely unknown. In addition, wintersweet belongs to magnoliids, and the phylogenetic position of magnoliids remains to be determined conclusively. Here, the whole genome of red flower wintersweet, a new wintersweet type, was sequenced and assembled with high quality. The genome comprised 11 super-scaffolds (chromosomes) with a total size of 737.03 Mb. Based on the analyses of the long branch attraction, incomplete lineage sorting, sparse taxon sampling, and other factors, we suggest that a bifurcating tree may not fully represent the complex early diversification of the angiosperms and that magnoliids are most likely sister to the eudicots. The wintersweet genome appears to have undergone two whole-genome duplication (WGD) events: a recent WGD event representing an independent event specific to the Calycanthaceae and an ancient WGD event shared by Laurales. By integrating genomic, transcriptomic, and metabolomic data, *CpANS1* and the transcription factor *CpMYB1* were found to play key roles in regulating tepal color development, whereas *CpMYB1* needs to form a complex with *bHLH* and *WD40* to fully perform its regulatory function. The present study not only provides novel insights into the evolution of magnoliids and the molecular mechanism for flower color development, but also lays the foundation for subsequent functional genomics study and molecular breeding of wintersweet.

Keywords: *Chimonanthus praecox*, genome, magnoliid evolution, flower color development, MYB transcription factor.

INTRODUCTION

The flowering plant clade Mesangiospermae mainly contains eudicots and monocots, accounting for 75 and 20% of angiosperms, respectively (Zeng et al., 2014). The magnoliids contain four orders, namely Canellales, Laurales, Magnoliales, and Piperales, and represent one of the largest clades outside monocots and eudicots (Cantino et al.,

2007; Heywood et al., 2007; Massoni et al., 2014, 2015). This clade contains some of the 'earliest angiosperms', which are distributed worldwide (Heywood et al., 2007; Zeng et al., 2014). Some Magnoliidae are economically important species, providing food for other organisms (Massoni et al., 2015). Despite the importance of magnoliids, their phylogenetic position relative to eudicots and

monocots has remained uncertain for decades. According to the different genomic components, such as plastomic inverted repeat regions, nuclear genes, mitochondrial genes, and plastid genes, the following three different points of view have been put forward: The magnoliids are (i) sister to the eudicots; (ii) sister to the monocots; or (iii) sister to the clade containing eudicots and monocots. Recently, the whole genomes of nine magnoliids have been sequenced, including stout camphor (*Cinnamomum kanehirae*) (Chaw et al., 2019), avocado (*Persea americana*) (Rendón-Anaya et al., 2019), *Litsea cubeba* (Chen, Li, et al., 2020), phoebe (*Phoebe bournei*) (Chen, Sun, et al., 2020), wintersweet (*Chimonanthus praecox*) (Shang et al., 2020), *Chimonanthus salicifolius* (Lv et al., 2020), *Liriodendron chinense* (Chen et al., 2019), black pepper (*Piper nigrum*) (Hu et al., 2019), and *Magnolia biondii* (Dong et al., 2021). These genomic data have been helpful for further understanding magnoliid evolution. However, the abovementioned nine independent studies did not produce identical results regarding phylogenetic placement, which may be due to different methods and species used. Therefore, the phylogenetic position of magnoliids still remains to be determined.

Calycanthaceae, including the genera *Chimonanthus*, *Calycanthus*, and *Idiospermum*, belongs to magnoliids (Chase et al., 2016) and form an independent branch of Laurales (Zhang et al., 2012). Among them, *Calycanthus chinensis* is mainly distributed in China, *Calycanthus floridus* and *Calycanthus occidentalis* mainly in North America, and *Idiospermum australiense* mainly in Australia. Some species of *Chimonanthus* are widely cultivated as ornamental plants because of their strong floral fragrance, and some others have medicinal value because of their alkaloids, sesquiterpenes, coumarins, and other secondary metabolites. *Chimonanthus praecox* originates in China and has been cultivated for more than 1000 years. Because of its high economic and ornamental value, *C. praecox* is also widely cultivated in America, Europe, and Japan as an ornamental plant.

Flower color plays a decisive role in the ornamental and economic value of most ornamental plants. Breeding new flower color varieties is one of the main goals of ornamental plant breeding. The overall color of wintersweet is mainly determined by the middle tepals, which are usually yellow or light yellow (Yang et al., 2018). According to the color of inner tepals, the cultivars of wintersweet are divided into three groups: the Concolor Group, the Intermedius Group, and the Patens Group. In the previous investigation of wintersweet germplasms, we found a new resource type of wintersweet named 'red flower wintersweet'; its inner and middle tepals are basically all red in early and full blossom. In order to distinguish them from the other three types, we classified the new wintersweet plants as the Rubrum Group. The discovery of this new

type provides resources for exploring the formation mechanism of flower color of wintersweet and provides a basis for the use of molecular marker-assisted breeding to accurately control flower color in the future.

In general, the types of *C. praecox* varieties mainly include *C. praecox* var. *concolor*, *C. praecox* var. *grandiflorus*, and *C. praecox* var. *intermedius*. Shang et al. (2020) reported the genome of *C. praecox* 'H29', which belongs to the Patens Group. In this study, we performed whole-genome sequencing of the newly released cultivar *C. praecox* 'Hongyun' (Figure S1), which represents the newly discovered Rubrum Group. Based on the PacBio Sequel II sequencing platform and Hi-C-assisted assembly, a high-quality chromosome-level genome was obtained. The genome sequence allowed us to redefine the evolutionary position of magnoliids by comparative analyses of the wintersweet genome with those of the other six magnoliids, monocots and eudicots, and other representative angiosperms. Finally, we explored the coloring mechanism of wintersweet based on genomic, transcriptomic, and metabolic data.

RESULTS

Genome sequencing, assembly, and annotation

The Rubrum wintersweet genome was sequenced by an Illumina Novaseq 6000 and PacBio SMRT to produce the V1.0 assembly. This initial genome assembly comprised 661 contigs covering 736.79 Mb, with a contig N50 of 8.13 Mb, which was approximately 96.17% of the predicted genome size by genome survey analysis (766.16 Mb). Using the Hi-C platform, we anchored 0.70 Gb of the 0.74 Gb contig sequences to 11 super-scaffolds (chromosomes), and the total length of scaffolds was 737.03 Mb (Table 1; Figure 1 and Tables S1–S3). The BUSCO analyses showed the completeness of the genome to be 95% (Table S4). We further evaluated this draft assembly using minimap2 by mapping continuous long reads to the

Table 1 Assembly features for the Rubrum wintersweet genome

Assembly feature	Value
Estimated genome size	766.16 Mb
Size of the assembly	737.03 Mb
Total length of contigs	736.79 Mb
Super-scaffold N50	66.25 Mb
Contig N50	8.13 Mb
Scaffold N50	63.91 Mb
GC content	36.58%
Chromosome number	11
Total length of super-scaffold	701.00 Mb
Total repeat size	486.64 Mb
Gene number	25,832
Average gene length	9965.35 bp

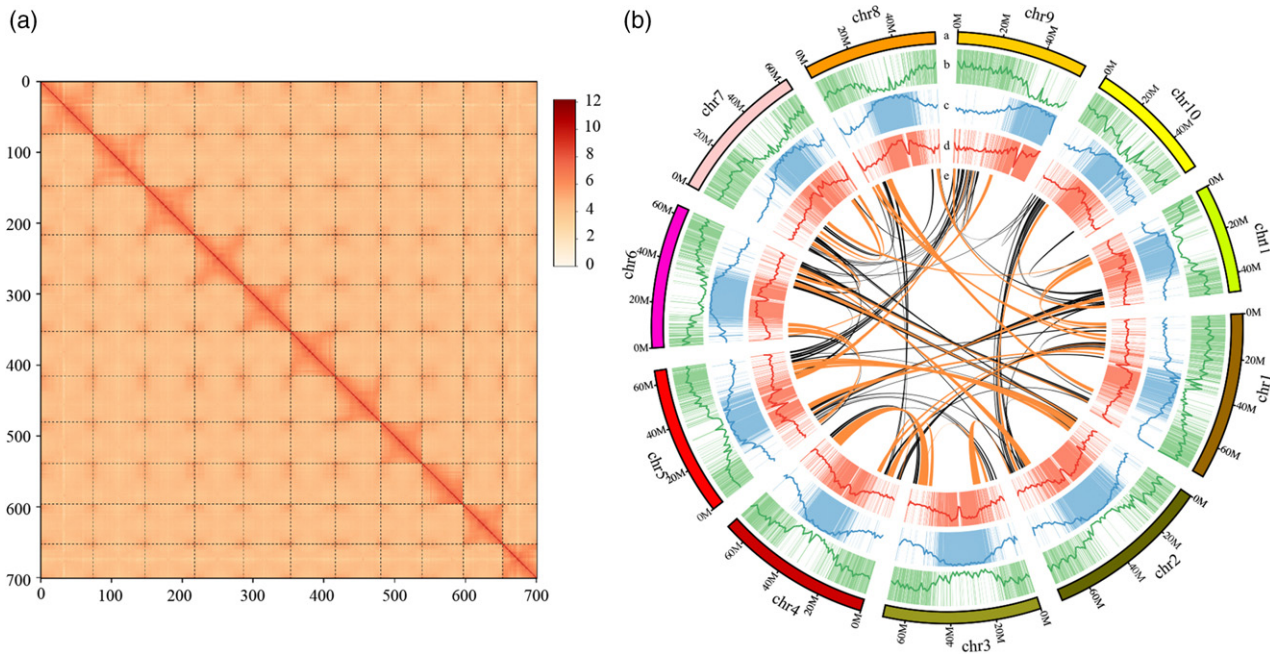


Figure 1. Hi-C-assisted assembly of *Chimonanthus praecox* 'Hongyun' V1.0 chromosomes.

(a) Hi-C interaction heatmap at a resolution of 200 kb. (b) The landscape of genome assembly and annotation of *C. praecox*. Tracks from outside to inside correspond to: a, assembled chromosomes; b, GC content; c, repeat density; d, gene density; and e, syntonic relationship.

genome assembly, and found the genome coverage was 99.94% and the mapping rate was 96.94% (Table S5).

According to homology-based and *de novo* predictions, as well as transcriptome data, 25 832 protein-coding gene models were identified and predicted with an average coding sequence (CDS) length of 1256 bp and an average gene length of 9965 bp. Among these, 24 756 (95.83%) of the predicted protein-coding genes could be mapped with functional annotations using public databases (Figure 1b and Table S6). These genes were unevenly distributed on each chromosome, and the gene density at chromosome arm ends was higher. Through a combination of homology-based searches and *de novo* prediction, 466.24 Mb repetitive sequences were identified, accounting for 63.26% of the genome size. Among the repetitive sequences, DNA transposons, long interspersed nuclear elements, short interspersed nuclear elements, and long terminal repeats accounted for 9.02, 7.13, 0.31, and 45.86% of the genome size, respectively (Figure 1b and Tables S7 and S8). We also identified 1723 non-coding RNAs (Table S9).

The phylogenetic placement of magnoliids among angiosperms

As mentioned in the introduction, the evolutionary status of magnoliids is still contentious. The Rubrum wintersweet genome provides additional data to explore this issue. Toward this end, a total of 70 single-copy orthologous gene sets were identified among 25 genomes (Tables S10

and S11), including seven magnoliids, eight eudicots, six monocots, two angiosperm basal group species, and two gymnosperms. A phylogenetic tree was first constructed by using the concatenated sequence alignments of full nucleotide sequences (covering phase 0, 1, and 2 loci) of 70 single-copy gene families (Figure S2a). To eliminate the influence of long branch attraction (LBA) (Bergsten, 2005; Qu et al., 2017), different phase loci with the exception of phase 2 loci (Ren et al., 2012; Sanderson et al., 2000), i.e., phase 0 and/or 1 loci, were used to reconstruct the concatenated tree (Figure 2a,b left and Figure S2c). Among the three trees produced, the two trees (based on phase 1 or phases 0 and 1) displayed reasonable results, with the seven magnoliids forming a magnoliid-specific sister group (Figure 2a,b left). Subsequently, a concatenated tree (Figure 2b right) was constructed using amino acid sequences of the 70 single-copy gene families. All three trees (Figure 2a,b left and Figure 2b right) showed that the magnoliids are sister groups to the eudicots with bootstrap support greater than 60% after their common ancestor diverged from monocots, which is consistent with the genome-based analyses of *C. kanehirae* (Chaw et al., 2019), *C. praecox* (Shang et al., 2020), and *C. salicifolius* (Lv et al., 2020).

The resolution of early-diverging branches within angiosperms may be confounded by incomplete lineage sorting (ILS). To avoid the influence of ILS, we constructed nucleic acid sequence-based and amino acid sequence-based coalescent trees using ASTRAL software. The nucleic

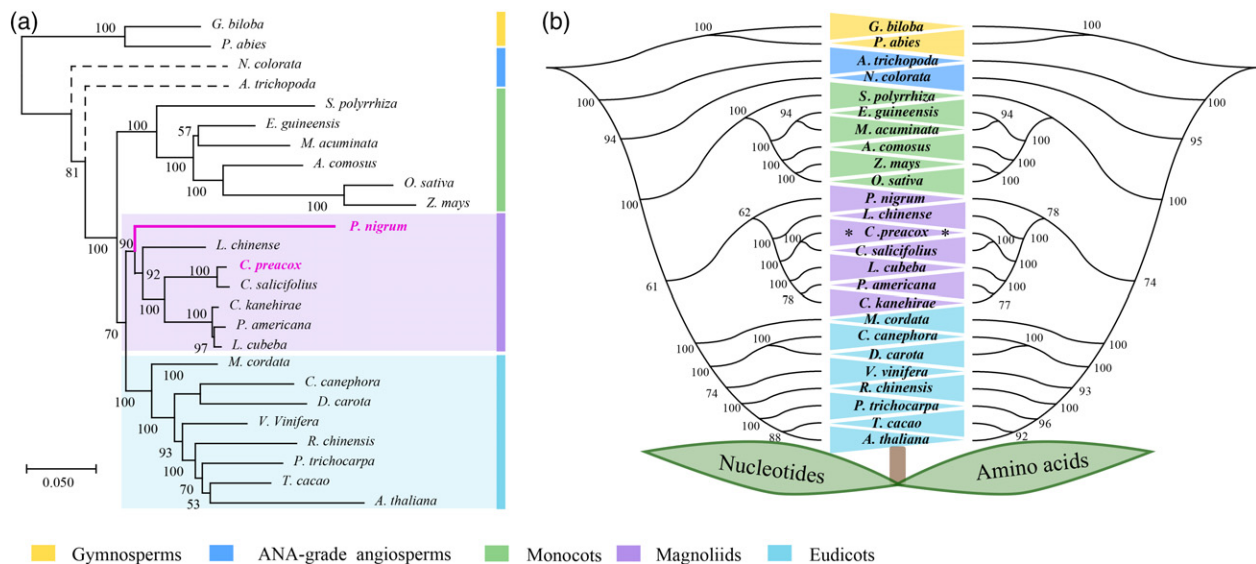


Figure 2. Phylogenetic relationship between *Chimonanthus praecox* and other plant species.

The phylogenetic trees were constructed using nucleotide and amino acid sequences of 70 single-copy gene families from 25 species. (a) The phylogenetic tree based on the concatenated sequence alignments of partial nucleotide sequences (covering phase 1 loci). (b) The phylogenetic trees based on the concatenated sequence alignments of partial nucleotide sequences (covering phase 0 and 1 loci; left) and amino acid sequences (right).

acid sequence-based coalescent tree (Figure S2d left) were in accordance with our concatenated trees, supporting a sister group relationship between eudicots and magnoliids; the amino acid sequence-based coalescent tree (Figure S2d right) supported a sister relationship between magnoliids and monocots.

Considering that sparse taxon sampling may affect the final result of evolutionary analysis, 47 transcriptomes and 76 genomes, including representatives of monocots, magnoliids, eudicots, ANA-grade, Chloranthales, Ceratophyllales, and a gymnosperm outgroup, were used to infer the phylogeny of major angiosperm lineages. Four low-copy nuclear (LCN) sets including 1082, 780, 597, and 99 genes were screened successively by orthologous gene set length, species coverage, and group monophyly (Tables S12–S15) and then applied to construction of 10 phylogenetic trees. The 10 trees included four nucleic acid sequence-based and four amino acid sequence-based coalescent trees with the four gene sets and one nucleic acid sequence-based and one amino acid sequence-based concatenated tree with the set of 99 genes (Figures S5–S14). Subsequently, we integrated two phylogenetic trees derived from five nucleic acid trees and five amino acid trees according to the topology inferred in our coalescent-based analysis of 1082 genes from 123 taxa (Figure 3 and Figure S4). Analyses of these tree topologies strongly supported the magnoliids as the sister of the eudicots with bootstrap support greater than 70% (the other two relationships among magnoliids, eudicots, and monocots were also present in some trees). Concerning the placement of ANA-grade, almost all the trees displayed the same

topology, with Amborella, Nymphaeales, and Austrobaileyales being successive sister groups to all other extant angiosperms.

Previous studies of the evolutionary position of magnoliids showed inconsistent results possibly due to different tree-building methods, the existence of ILS, the limitation of taxon sampling, and the rapid differentiation of magnoliids in the early stages of evolution. This study fully considered various factors that may have caused errors in the evolutionary positioning of the magnoliids. Our data suggest that the bifurcating tree based-approaches might not be able to fully demonstrate the early diversification of the angiosperms. We suggest that magnoliids are most likely sister to eudicots, which was supported by the results from several methods.

Whole-genome duplications in wintersweet

From the distribution of synonymous substitutions per synonymous site (K_s), the *C. praecox* genome showed two peaks with whole-genome duplication (WGD) events, including an ancient event with a K_s value of 0.9–1 and a recent event with a K_s value of 0.6 (Figure 4a). Two similar peaks at K_s values were also detected in the other five Calycanthaceae species (*C. salicifolius*, *C. chinensis*, *C. floridus*, *C. occidentalis*, and *I. australiense*) (Figure S15a–e), suggesting that two WGD events are shared by all Calycanthaceae species. The results of the K_s distributions were consistent with the genetic distance between *C. praecox* and other species, including Calycanthaceae (*C. chinensis*, *C. floridus*, *C. occidentalis*, *C. salicifolius*, and *I. australiense*), magnoliids (*Cassytha filiformis*, *C. kanehirae*,

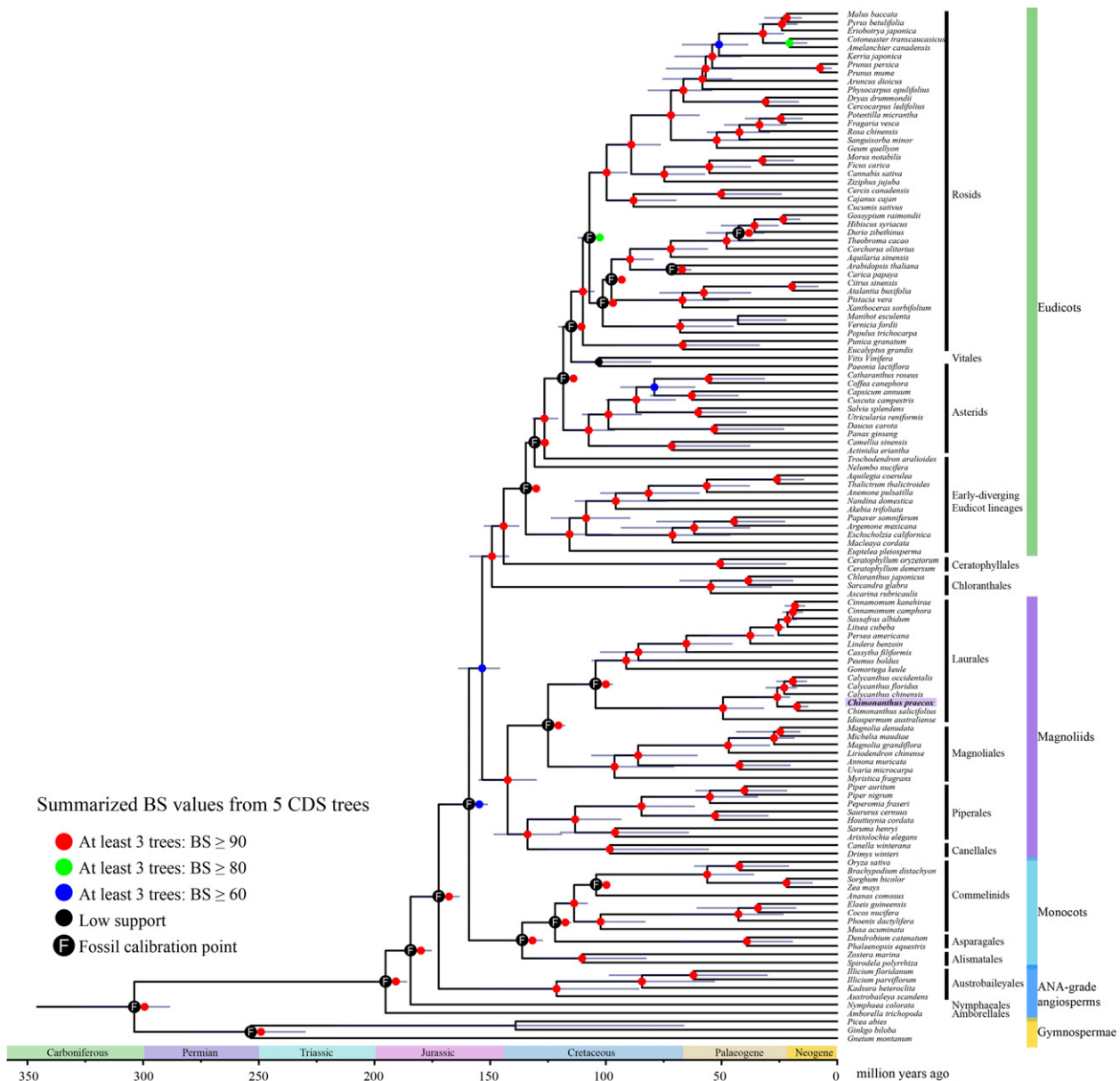


Figure 3. Summary of phylogeny and timescale of 123 plant species based on nucleotide sequences of selected low-copy nuclear ortholog groups. Blue bars represent 95% credibility intervals of the estimated divergence time.

P. americana, *L. cubeba*, *Lindera benzoin*, *Sassafras albidum*, *Gomortega keule*, *Peumus boldus*, *L. chinense*, and *P. nigrum*), and the other four species (*Macleaya cordata*, grape [*Vitis vinifera*], *Amborella trichopoda*, and *Ginkgo biloba*) (Figure 4c–e and Figure S15q,r). In addition, the relationship between WGD events and the species differentiation of Laurales and magnoliids was explained by comparing K_s values between *C. praecox* and these species with a K_s value of paralogous gene pairs in these species (Tables S16 and S17). Interestingly, the K_s value between *C. praecox* and *L. chinense* (approximately 0.85, Figure 4d and Table S17) was slightly smaller than that of the

C. praecox ancient WGD event (approximately 0.92, Figure 4a and Table S16), but larger than that of the *L. chinense* WGD event (approximately 0.68, Figure 4b and Table S16). This would lead to the opposite conclusion in determining whether Magnoliaceae and Calycanthaceae share the same WGD event. For this reason, we characterized the relationship between the occurrence of WGD and the species differentiation of Calycanthaceae, Lauraceae, Laurales, Magnoliales, and magnoliids by analyzing K_s values between synteny homoeologs within or between wintersweet and other Magnoliids (Figure 4a–e and Figure S15) according to the methods described for the

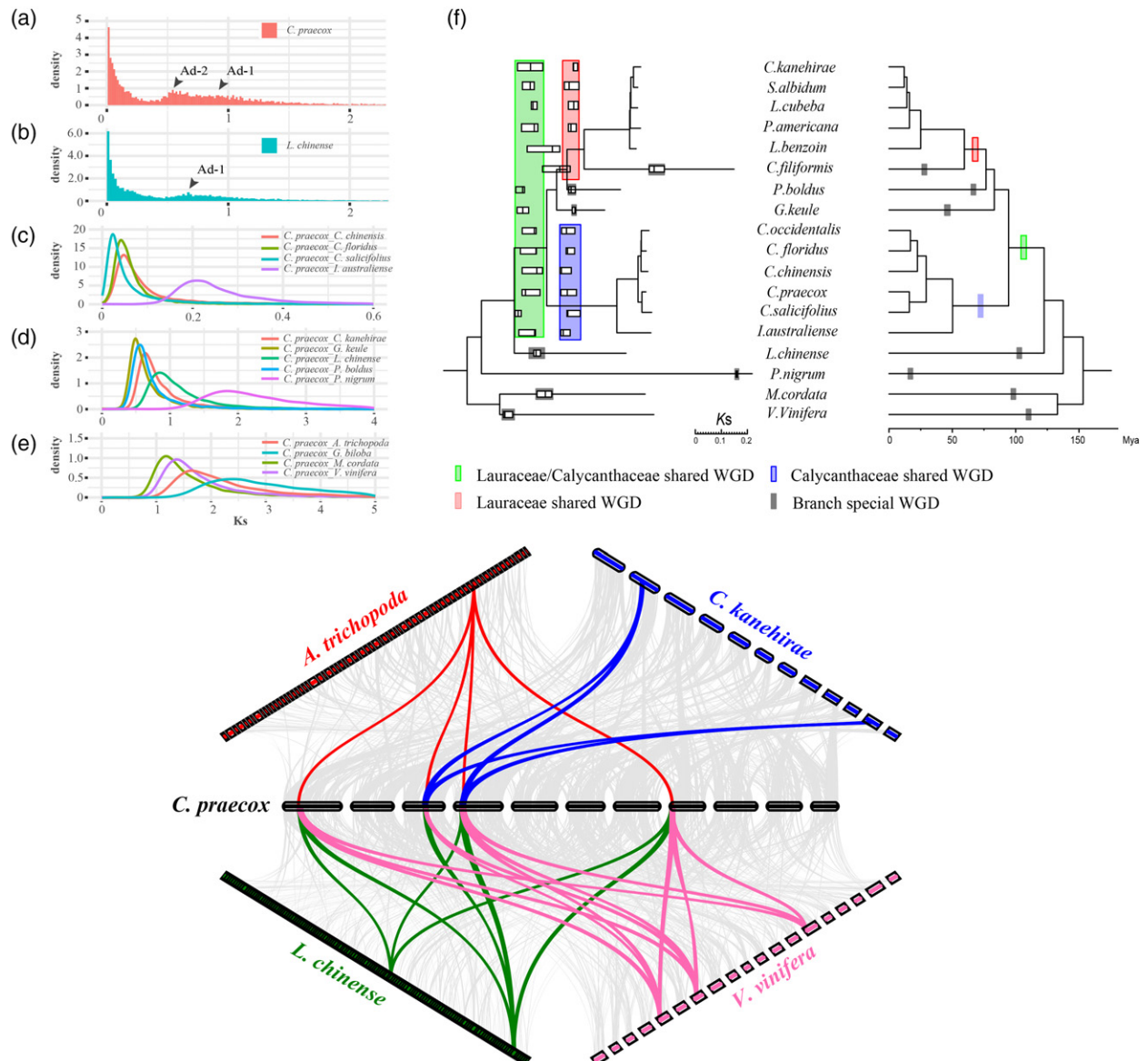


Figure 4. Whole-genome duplications of *Chimonanthus praecox*.

(a) Synonymous substitution level (K_s) distribution of paralogous gene pairs in *C. praecox*, showing two ancient duplication events (Ad) occurred. (b) Synonymous substitution level (K_s) distribution of paralogous gene pairs in *L. chinense*, showing one ancient duplicate event (Ad) occurred. (c–e) K_s distribution of orthologous gene pairs between *C. praecox* and selected Calycanthaceae, magnoliids, and four other species. (f) The phylogeny of Lauraceae and Calycanthaceae based on K_s values. (g) Intragenomic synteny between *C. praecox* (11 chromosomes) and each of *A. trichopoda* (72 scaffolds), *C. kanehirae* (12 chromosomes), *L. chinense* (142 scaffolds), and *Vitis vinifera* (19 chromosomes).

Litsea genome (Chen, Li, et al., 2020). As shown in Figure 4f, we found that the common ancestor of Laurales underwent a diploidization and then differentiated into the Calycanthaceae and Lauraceae, etc., and then experienced a unique WGD event, except for *C. filiformis*. Piperaleae (*L. chinense*), Magnoliales (*P. nigrum*), and Laurales, which did not experience a common WGD event but experienced a specific WGD event after their differentiation. Additionally, the phylogenetic data suggest that the ancient WGD event of Laurales and the WGD event of *Liriodendron* were

relatively close, but they were independent WGD events after differentiation.

The synteny analysis between *C. praecox* and *C. kanehirae*, *L. chinense*, *V. Vinifera*, and *A. trichopoda* also supported two WGD events in *C. praecox* (Figure 4g) and revealed one orthologous region in *A. trichopoda* corresponding to four paralogous segments in *C. praecox*. In addition, the comparisons of *C. praecox* with *C. kanehirae*, *L. chinense*, and *V. Vinifera* presented a specific synteny pattern of 2:2, 4:2, and 4:3, respectively. The detailed

comparisons of *C. praecox* with *C. salicifolius* presented a specific synteny pattern of 1:1 (Figure S16). The synteny analysis showed that *C. salicifolius* experienced two WGD events, which were shared with *C. praecox*; *Cinnamomum* also experienced two WGD events, the ancient event shared with *Chimonanthus*, and then they respectively experienced one WGD event after differentiation; *Liriodendron* experienced a WGD event after differentiation. These results were in accordance with the K_s analyses as well.

Taken together, these results suggest that the younger of the two WGD events in *C. praecox* was likely to be shared among Calycanthaceae, and that was an independent, Calycanthaceae-specific event. We analyzed the gene trees containing at least one paralogous pair from synteny regions from one of the five Laurales genomes, placed the WGD of *C. praecox* and the WGDs of *C. salicifolius*, *C. kanhiraie*, *P. americana*, and *L. cubeba* on the Laurales stem branch, and also confirmed additional evidence for the common Laurales WGD. These results further provide strong evidence for a WGD shared by all Laurales, which

might be only slightly older than their earliest differences and is likely to be correlated with Laurales divergence. In addition, the results also show that Laurales and Magnoliales did not share a WGD event before they diverged from magnoliids.

Cyanidin glycosides are the main metabolites of the red tepals in wintersweet

Tepal color is the major characteristic of wintersweet. For example, both of the middle and inner tepals of Concolor wintersweet are yellow; the middle tepals of Patens wintersweet are yellow and the inner tepals are red; and both of the middle and inner tepals of Rubrum wintersweet are red. To gain insight into the difference of wintersweet flower color, we detected the flavonoid-targeted metabolites using UPLC-MS/MS in the middle and inner tepals of three types of flowers at the early flowering stage, including Concolor wintersweet, Patens wintersweet, and Rubrum wintersweet (Figure 5a). Subsequently, we detected seven different anthocyanins, including three

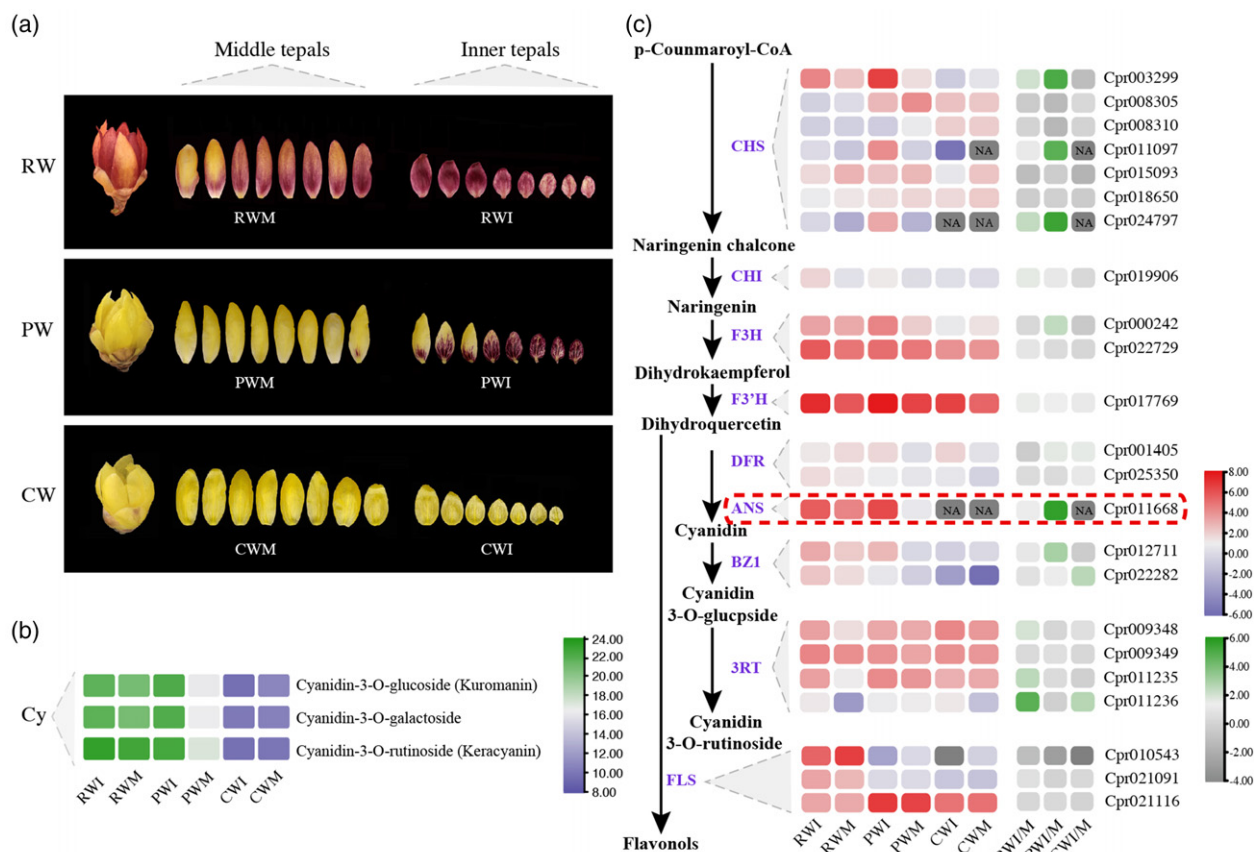


Figure 5. Identification of key candidate genes in pigment biosynthesis of *Chimonanthus praecox*.

(a) Phenotype of flowers of Rubrum, Patens, and Concolor wintersweet (RW, PW, and CW) at the early flowering stage. (b) The relative anthocyanin content of inner (I) and middle (M) tepals of RW, PW, and CW, respectively. (c) The expression profiles of structural genes involved in the anthocyanin biosynthesis pathway based on the FPKM values. 'RWI/M', 'PWI/M', and 'CWI/M' represent the specific value of inner and middle tepals' FPKM values of RW, PW, and CW, respectively. NA, not available/missing values. All data were $\log_2$ -transformed before the creation of heatmaps.

cyanidins (Cy), two delphinidins (Dp), one peonidin (Pn), and one petunidin (Pt) (Table S18 and File S1). Compared with other anthocyanin metabolites, the contents of three types of cyanidin glycosides (i.e., cyanidin-3-O-galactoside, cyanidin-3-O-glucoside, and cyanidin-3-O-rutinoside) in the red tepals were significantly higher than those in the yellow tepals (Figure 5b and Table S18). Taken together, these results suggest that cyanidin glycosides cause the difference in color of the wintersweet flowers.

ANS1 and transcription factors possibly play key roles in the flower color development of wintersweet

To further understand the molecular mechanism of red tepal formation in wintersweet, we performed transcriptome sequencing using the same tissues as for the metabolomics analysis. According to the functional annotation of the whole genome, we further annotated all genes involved in flavonoid (ko00941) and anthocyanin (ko00942) biosynthesis. Among them, *CpANS1* (Cpr011668) was highly expressed in red tepals, but almost negligibly in yellow tepals (Figure 5c and Table S19). Combined with the metabolomics and transcriptomics approaches, *CpANS1* was considered a pivotal structural gene responsible for the formation of red color in wintersweet.

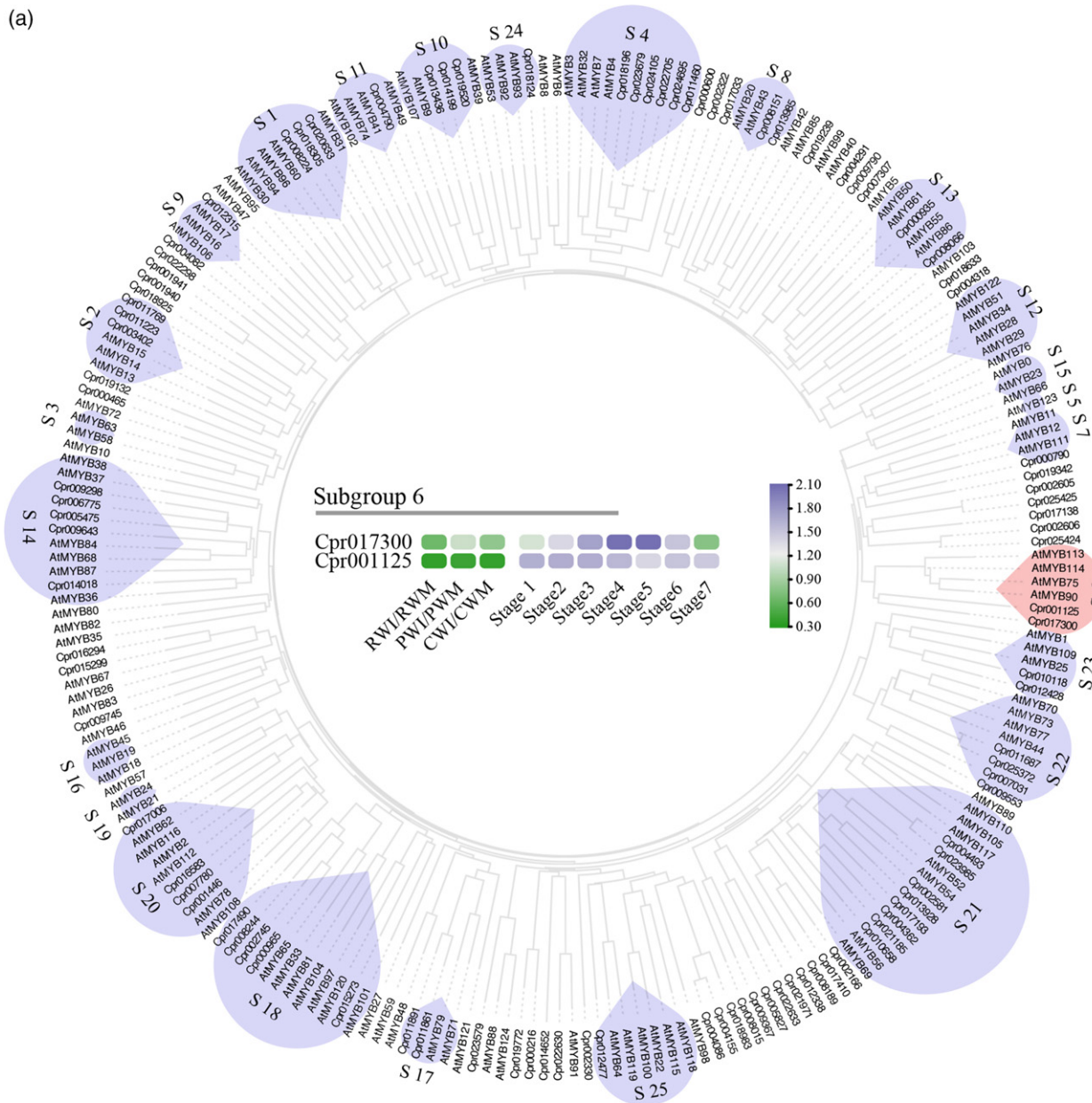
To explore the cause of high expression of *CpANS1* in red tepals of wintersweet, we isolated and cloned the CDSs of *CpANS1* and the corresponding promoter fragment from three different varieties (Figure S17a,c; Tables S20 and S21). Sequencing results showed that there was no difference in the core promoter regions and translated amino acid sequences of the *CpANS1* gene among these three varieties. These results suggest that the difference in promoter sequences or translated amino acid sequences of *CpANS1* was not the cause of flower color difference among these three varieties. Indeed, the early biosynthetic genes (EBGs), including *FLS1*, *F3'H*, *F3H*, *CHI*, and *CHS*, can lead to flavonol biosynthesis; the late biosynthetic genes (LBGs), including *ANR*, *ANS*, and *DFR*, can catalyze both anthocyanin and proanthocyanidin (PA) biosynthesis in *Arabidopsis thaliana*. EBGs are positively regulated by subgroup 7 R2R3-MYBs, whereas LBGs are activated by subgroup 6 R2R3-MYBs or AtMYB123 when leading to the biosynthesis of anthocyanins and PAs, respectively (Cao, Li, et al., 2020; Dubos et al., 2010). In our study, two methods were used in combination to determine the upstream regulatory genes of *CpANS1*. Firstly, we identified 99 R2R3-MYB transcription factors at the genomic level (Table S22). The maximum likelihood (ML) tree divided these MYB transcription factors into 25 different subgroups using 125 MYBs from *A. thaliana* as templates. Two MYBs (*Cpr017300* and *Cpr001125*) of the S6 subgroup were identified by phylogenetic tree analysis (Figure 6a and Table S23). Subsequently, weighted gene co-expression network analysis (WGCNA) was carried out based on the

transcriptome data of seven tepal samples (Figure 6b). Three MYBs were identified and shown to be co-expressed with *CpANS1* (Table S24). Among the three MYBs, *Cpr008224*, and *Cpr020633* were clustered with *A. thaliana* subgroup 1 MYBs (Figure 6a). Combined with phylogenetic and WGCNA analyses, *Cpr017300* (annotated as *CpMYB1*) might encode a key transcription factor regulating the expression of *CpANS1*. The correlation analysis of different comparison groups between red and yellow tepals (Table S25) further confirmed that *CpANS1* expression was positively correlated with cyanidin glycosides and that *CpMYB1* might be positively correlated with *CpANS1* (Figure S18).

CpMYB1 might regulate flower color difference among different wintersweet groups

Among the three varieties of wintersweet, RWM, RWI, and PWI tepals are red and CWM, CWI, and PWM tepals are yellow (Figure 5a). Significantly, a deletion of four bases (AAAG) of *CpMYB1* was present in transcripts found in both inner (CWI) and middle (CWM) tepals of Concolor wintersweet in comparison with Rubrum wintersweet (RW) and Patens wintersweet (PW), as determined by cloning the CDSs of *CpMYB1* from three different varieties (Figure 7a, Figure S17b; Tables S20 and S21). For CDSs of *CpMYB1*, the number of 'GGA' repeats after 'AAAG' varied in different red tepal tissues (Figure 7a). The sequences from transcriptome data were also verified by PCR amplification. The deletion of 'AAAG' in the CDS of *CpMYB1* in Concolor wintersweet causes a frameshift mutation, which eventually leads to early termination (62 bp) of *CpMYB1* (Figure 7a). In general, the activation or repression domain of R2R3-MYB transcription factors is usually located at the C-terminus (Cao, Li, et al., 2020; Dubos et al., 2010). Therefore, the frameshift mutation of *CpMYB1* may lead to the loss of its regulatory function and eventually cause termination of the anthocyanin metabolic pathway in Concolor wintersweet even if it is highly expressed in the inner tepals. In Patens and Rubrum wintersweets, by contrast, *CpMYB1* exhibits a higher expression level in red tepals than yellow tepals (Table S23), indicating that the *CpMYB1* expression level is directly related to the color of the tepals. Therefore, we suggest that *CpMYB1* might regulate flower color difference among different wintersweet groups: its deletion causes the red color loss in Concolor wintersweet, while its expression level determines the tepal color in Patens and Rubrum wintersweets. In addition, we also found that there was no difference in the sequence of the *CpMYB1* core promoter among the three varieties by cloning the promoter sequence (Figure S17c; Tables S20, and S21), which indicates that other regulatory factors instead of the promoter may cause the specific expression of *CpMYB1*.

(a)



(b)

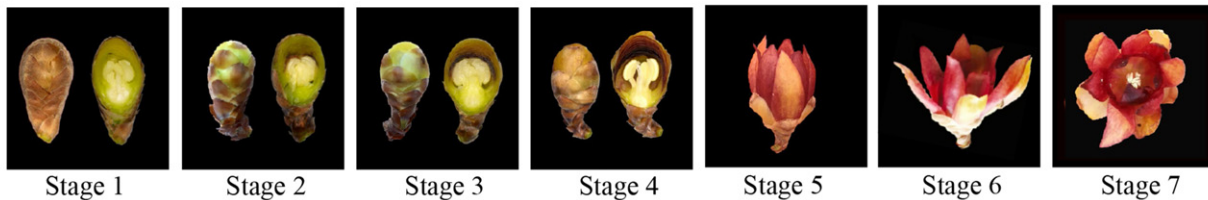
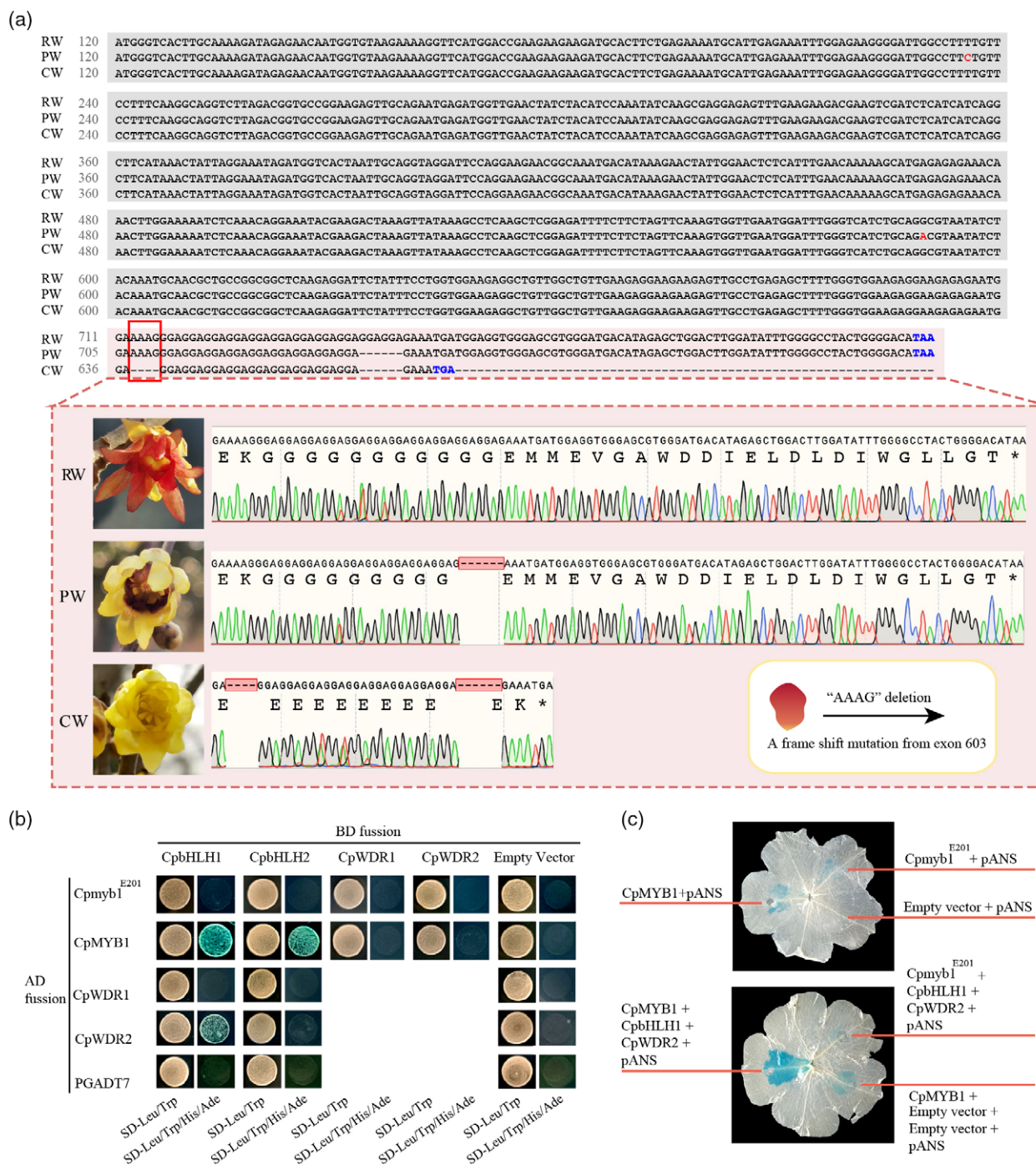


Figure 6. Identification of anthocyanin-related R2R3-MYB candidates of *Chimonanthus praecox*.

(a) Phylogenetic tree of R2R3-MYB transcription factors of *C. praecox* (99 members) and *Arabidopsis thaliana* (125 members). The specific values of inner and middle tepals of three varieties based on the FPKM values and the FPKM values of all subgroup 6 *CpMYBs* in seven developmental stages are shown in the center of the tree. 'RWI/M', 'PWI/M', and 'CWI/M' represent the specific values of inner and middle tepals' FPKM values of RW, PW, and CW, respectively. The FPKM values were log₂-transformed before the creation of heatmaps. (b) The phenotype of flowers of Rubrum wintersweet in seven developmental stages. Stage 1, sprout period of scale bud; Stage 2, early period of green bud; Stage 3, late period of green bud; Stage 4, yellow bud period; Stage 5, early flowering period; Stage 6, full flowering period; Stage 7 wither period.



(a) The complete coding sequence alignment and the protein sequences in the C-termini of CpMYB1 from Rubrum (RW), Patens (PW), and Concolor (CW) winter-sweet tepals. (b) Yeast two-hybrid (Y2H) analysis of the physical interaction between (i) CpMYB1 and (ii) CpbHLH1, CpbHLH2, CpWDR1, and CpWDR2. (c) GUS activity assay in petals of white *Petunia* × *hybrida*.

(Cpr013636), and CpWDR2 (Cpr014700) (Table S26). A yeast two-hybrid experiment showed that CpMYB1 can interact with CpbHLH1 and CpbHLH2, but cannot interact with CpWDR1 and CpWDR2. Further results indicated that there is also an interaction between CpbHLH1 and CpWDR2 (Figure 7b). Therefore, an MBW complex can be formed between CpbHLH1, CpWDR2, and CpMYB1.

To confirm the activation activity of CpMYB1 and its MBW complex on CpANS1, we conducted overexpression experiments in petunia W59 $\times$ axi petals. The GUS staining results show that the CpMYB1-driven MBW complex has a strong activating activity on the CpANS1 promoter, while Cpmbyb1^{E201} (a synonymous mutation at amino acid 201, and with AAAG bases deleted from site 603 of the CDS, resulting in a frameshift mutation) loses the ability to activate the CpANS1 promoter (Figure 7c). At the same time, we also conducted tobacco (*Nicotiana benthamiana*) overexpression experiments to verify the function of CpMYB1. The results showed that neither CpMYB1 nor Cpmbyb1^{E201} could deepen the color of tobacco leaves and calli (Figure S19), indicating that CpMYB1 may need to form a complex (MBW) with bHLH and WD40 in *C. praecox* to exercise its regulatory function.

DISCUSSION

Wintersweet is a famous ornamental plant with high economic value and cultural connotation. In the present study, we described a high-quality chromosome-scale genome assembly for Rubrum wintersweet *C. praecox* 'Hongyun' with a scaffold N50 of 63.91 Mb and a contig N50 of 8.13 Mb. The Rubrum wintersweet genome sequence adds to the growing body of genome information for the magnoliids and provides important resources for future evolutionary studies, investigations of the coloration mechanism, and molecular breeding.

The members of the magnoliids have different phylogenetic positions in the four different versions of the Angiosperm Phylogeny Group (APG) system (Baas et al., 2000; Chase et al., 2016; Haston et al., 2007, 2009). For example, magnoliids are sister to monocots in APG II, but sister to both monocots and eudicots in APG I, III, and IV. Up to now, a total of nine genomes of magnoliids have been sequenced and published. Among them, Chaw et al. (2019), Lv et al. (2020), and Shang et al. (2020) supported the notion that magnoliids are sisters to eudicots; meanwhile, Chen et al. (2019), Hu et al. (2019), Rendón-Anaya et al. (2019), Chen, Sun, et al. (2020), and Dong et al. (2021) supported the notion that magnoliids are sisters to monocots and eudicots. The differences in the methods of constructing trees may have led to the different results with respect to the evolutionary position of magnoliids (Chen, Sun, et al. 2020; Chen, Li, et al., 2020; Rendón-Anaya et al., 2019). The evolutionary relationship among the eudicots, monocots, and magnoliids thus seems

unresolved. Many factors may contribute to the topological differences, such as the number of retrieved orthologs, the LBA, the possible ILS, and the taxon sample size (Bergsten, 2005; Wiens, 2005; Zhang et al., 2020).

LBA is a major obstacle to phylogenetic reconstruction, which can cause distantly related lineages to be incorrectly inferred as closely related. For example, the phylogenetic relationships among the Hesperocyparis/Callitropsis/Xanthocyparis, Cupressus, and Juniperus sub-branches of Cupressoideae are controversial. Qu et al. (2017) found that both Juniperus and Cupressus had much longer stem branches than Hesperocyparis/Callitropsis/Xanthocyparis, so their sister relationships may be attributed to LBA. In this study, LBA occurred in the concatenated tree using phase 0, 1, and 2 loci, which classified *P. nigrum* as a monocot and supported the notion that magnoliids are sister to monocots (Figure S2a). To confirm this observation, we rebuilt the concatenated phylogeny excluding *P. nigrum*, and the newly generated trees showed that magnoliids were clustered together as sister to eudicots (Figure S2b). To solve the problem of LBA, the concatenated trees using phase 0, phase 1, and phase 0 and 1 loci of orthologous genes were used to reconstruct the phylogeny.

In the plant kingdom, many species are characterized by century-long fertile life spans, limited interspecific divergence, and large effective population sizes. These factors seem to have created a significant evolutionary network, which is deeply influenced by ILS. ILS is a result of polymorphic alleles in the ancestral populations (Chen, Li, et al., 2020). We used the Q value in ASTRAL to display the percentages of gene trees in support of three different branching scenarios for magnoliids, eudicots, and monocots. The q1, q2, and q3 values in the ASTRAL nucleotide tree were almost the same, and the q1 and q2 values were very close in the ASTRAL amino acid tree (Figure S3), which indicates that the ASTRAL tree might produce different results, due to different species selection, using different orthologous gene sets in previous studies for the evolutionary relationship among the three branches.

Adequate sampling of taxa, especially small plant lineages such as Chloranthales and Ceratophyllales in the angiosperm clade, plays an important role in obtaining distinguishable phylogeny. To eliminate the possible one-sided results caused by small sample size, multiple phylogenetic trees were constructed using transcriptomic and genomic data from 123 widely representative species, including representatives of the five major groups (eudicots, monocots, magnoliids, Chloranthales, and Ceratophyllales) in Mesangiospermae, ANA-grade, and a gymnosperm outgroup according to different subsets. Although many phylogenetic trees supported the notion that magnoliids are sister to eudicots, they also indicated that the rapid divergence of early mesangiosperms was

complicated. For example, the evolutionary positions of Chloranthales and Ceratophyllales in 10 phylogenetic trees were also inconsistent. Yang et al. (2020) also suggested that a fully bifurcating tree may be an inadequate representation of the early radiation of angiosperms.

It was suggested in their previously reported WGD results that one WGD event had occurred in the ancestry of Magnoliales (Magnoliaceae) and Laurales (Lauraceae) for the *Cinnamomum*, *Liriodendron*, *Litsea*, *Chimonanthus salicifolius*, and *Phoebe* genomes, respectively (Chaw et al., 2019; Chen et al., 2019; Chen, Sun, et al., 2020; Chen, Li, et al., 2020; Lv et al., 2020). However, Shang et al. (2020) concluded that the ancient wintersweet WGD event occurred after the divergence of Calycanthaceae and Magnoliaceae. In our study, based on orthologous divergences, K_s plots, and synteny analyses, the available evidence confirms that Piperales (*Piper*) and Magnoliales (*Liriodendron*) experienced one WGD event, not shared with Laurales. Two rounds of WGD were inferred and placed in the *C. praecox* genome, and the ancient WGD event was shared by Laurales, but Calycanthaceae, Lauraceae, *Gomortega keule*, and *Peumus bolds* experienced a specific WGD event (Figure 4f).

Anthocyanin glycosides are water-soluble flavonoid compounds, widely present in the roots, stems, leaves, flowers, fruits, and seeds of plants, and contribute to the different colors of these tissues and organs (Veitch and Grayer, 2011). Previous studies have shown that the main flavonols and anthocyanins in red inner tepals of wintersweet are cyanidin-3-*O*-glucoside, cyanidin-3-*O*-rutinoside; quercetin-3-*O*-rutinoside, kaempferol-3-*O*-rutinoside, and quercetin aglycone (Yang et al., 2018). Generally, the anthocyanins of magenta, red-purple, red, scarlet, and pink flowers are pelargonidin and/or cyanidin with or without peonidin (Iwashina, 2015). In our study, we also found contents of three types of cyanidin in the red tepals to be significantly higher than those in the yellow tepals (Figure 5b). These data indicate that cyanidin glycosides are the main reason for the red color of wintersweet.

The anthocyanin biosynthesis pathway has been widely studied in plants (Misyura et al., 2013). Anthocyanins are produced by a series of enzymes encoded by structural genes, including *CHS*, *CHI*, *F3H*, *F3'H*, *F3'5'H*, *DFR*, *ANS*, and *3GT*, and then transported into vacuoles for storage through various modifications (Cao, Li, et al., 2020; Misyura et al., 2013). Among them, *ANS* encodes a key enzyme in the anthocyanin biosynthetic pathway which catalyzes the conversion of colorless anthocyanins to colored anthocyanins. *ANS1* of *Gentiana triflora* has lost four nucleotides in the second exon, which leads to an early stop codon and inactivation of the gene (Nakatsuka et al., 2012). *ANS* in *Mirabilis jalapa* (*MjANS*) exhibits a lack of enzyme activity even though it is highly expressed in the

flowers, resulting in a lack of anthocyanins (Polturak et al., 2018). The expression level of *ANS* plays an important role in the synthesis of anthocyanins in the petals of *Malus* spp., the fruits of *Duchesnea indica*, the leaves of *Perilla frutescens*, and the inner tepals of *C. praecox* (Debes et al., 2011; Gong et al., 1997; Yang et al., 2018; Zhang et al., 2015). Our study combined metabolome and transcriptome data analyses and found that *CpANS1* might be the key gene that determines the color of wintersweet tepals. In contrast, *CpANS2* (Table S19), the corresponding duplicate of *CpANS1*, was slightly expressed in red tepals, indicating that *CpANS2* may have undergone neofunctionalization or subfunctionalization. The structural genes involved in the anthocyanin biosynthetic pathway can be regulated by some well-known transcription factors or complexes, such as MYB and MBW (Gu et al., 2019). Members from the MYB S6 subgroup mainly regulate anthocyanin synthesis (Allan and Espley, 2018; Cao, Li, et al., 2020; Dubos et al., 2010; Feller et al., 2011). In our study, CpMYB1 from the S6 subgroup or the MBW transcription complex formed by CpMYB1, bHLH, and WD40 may be involved in the regulation of *CpANS1* expression. Indeed, yeast two-hybrid experiments confirmed that CpMYB1, bHLH1, and WDR2 can form an MBW complex (Figure 7b). Transgenic tobacco experiments also indicated that CpMYB1 may need to rely on bHLH1 and WDR2 to form a complex to regulate anthocyanin metabolism.

Mutations in the CDS or promoter of *MYB*, the regulatory gene of the anthocyanin metabolism pathway, can affect its own activity, thus affecting the expression of downstream genes. For example, a special single-nucleotide polymorphism (SNP) (W12S) in the CDS of *Fve-MYB10* in strawberry (*Fragaria vesca*) is mutated from G to C, and the corresponding amino acid in this SNP is mutated from Trp to Ser, which causes this gene to lose the function of promoting anthocyanin accumulation and thereby causes the strawberry fruit color to change from red to yellow (Hawkins et al., 2016). The white color of grapes is caused by the simultaneous mutation of *Vv-MYBA1* and *VvMYBA2* (Walker et al., 2007). The red color of apple flesh is caused by a short tandem repeat of the *MdMYB10* promoter (Espley et al., 2009). The microsatellite sequence of the *McMYB10* promoter in *Malus crabapple* can significantly upregulate the expression of the gene, thus promoting the synthesis of anthocyanins (Tian et al., 2015). In this study, we compared *CpMYB1* among different varieties of inner and middle tepals and identified a deletion of four bases (AAAG) in Concolor wintersweet and a change in GGA repeat number in Patens wintersweet and Rubrum wintersweet. Meanwhile, we found that there was no difference in *CpMYB1* core promoter sequence among the three varieties (Figure S17; Tables S20 and S21). Therefore, we suggest that the CDS difference and specific expression of *CpMYB1* might determine the flower

color difference among different wintersweet groups. Further experiments showed that the MBW complex formed by CpMYB1 has activating activity on the *CpANS1* promoter, while Cpmyb1^{E201} has no activating activity on the *CpANS1* promoter (Figure 7c). Taken together, *CpMYB1* is a key gene that regulates the metabolic pathway of anthocyanins and determines the flower color, requiring the participation of CpbHLH1 and CpWDR2.

CONCLUSIONS

In this study, we have sequenced and assembled a high-quality genome of Rubrum wintersweet, a new wintersweet type. The genome sequence presented here provides a valuable resource for genome editing and molecular marker-assisted breeding and provides insight into the evolution of the magnoliids. The integration of multi-omics data elucidated the molecular mechanisms underlying Rubrum wintersweet's intriguing flower color development, which can now confidently be attributed to *CpANS1* and *CpMYB1*. At the same time, our experimental results also confirmed that CPMYB1 needs to form a complex with bHLH and WD40 to fully perform its regulatory function.

EXPERIMENTAL PROCEDURES

Plant materials

In the genome project, plant materials were collected from a 7-year-old red flower wintersweet *C. praecox* 'Hongyun' in Yanling experimental base of Henan Academy of Forestry, China. Young leaves were collected in spring for whole-genome sequencing. Leaves, roots, stems, flowers, and fruits of *C. praecox* 'Hongyun' were sampled for RNA sequencing. Different-colored tepals from three wintersweet groups were used for transcriptomic and metabolomic analyses. All the harvested tissues were immediately flash frozen and then stored at -80°C .

Library construction and sequencing

The cetyltrimethylammonium bromide method was used to extract total DNA from the leaves of *C. praecox* 'Hongyun'. First, for Illumina sequencing, the libraries were subjected to 150-bp paired-end sequencing on an Illumina Novaseq 6000 platform. Fastp was used to process the short reads (Chen et al., 2018). In total approximately 68.76 Gb of raw sequencing data were generated corresponding to roughly 90 times of the assembled genome. Second, at least 10 μg of sheared DNA was used to construct SMRTbell template libraries of 30 kb for PacBio SMRT (Sequel II) sequencing. Totally, we obtained 4.9 million PacBio long reads which generated approximately 149.5 Gb (200 $\times$ coverage) after data pre-processing and filtering. Third, one Hi-C library was prepared on an Illumina Novaseq 6000 platform to generate 150-bp read pairs using a previously described method (Lieberman-Aiden et al., 2009).

Genome assembly and assessment

We performed the contig-level assembly on full PacBio long reads by Meact2 software. Then the draft assembly was polished by Arrow (Chin et al., 2013). To further improve the accuracy of

assembly, we recruited Illumina short reads for further polishing by Pilon (v1.22). BUSCO4 was used to measure genome continuity and completeness (Parra et al., 2007; Simão et al., 2015). The paired-end reads from Hi-C were uniquely mapped onto the draft assembly contigs, which were grouped into chromosomes and scaffolds.

Genome annotation

Gene prediction was carried out using three strategies: RNA-seq data, homology information, and *de novo* prediction. To obtain gene functions, the predicted protein sequences of wintersweet were aligned to proteins annotated in TrEMBL, KEGG, and SwissProt using BLASTP software with an *E*-value threshold of 1×10^{-5} (Boeckmann et al., 2003; Lavigne et al., 2008). For identifying motifs and domains, we used InterProScan to search the Gene Ontology and InterPro databases (Gene Ontology Consortium, 2019; Jones et al., 2014). Infernal software was used to predict miRNA, snRNA, and rRNA genes by searching the Pfam database (Chan and Lowe, 2019; Mistry et al., 2020). RNAmmer was used to identify the rRNA subunits according to hidden Markov models (Lagesen et al., 2007). tRNAscan-SE was used to find the tRNA genes by applying eukaryote parameters (Lowe and Eddy, 1997). The transposable elements and repetitive sequences were predicted by *de novo* and homology-based approaches using previously described methods (Zhang et al. 2019).

Evolution and evolutionary rate estimation

By running OrthoMCL v2.0.9 program, we identified orthologous gene groups of wintersweet and 24 other species (Li et al., 2003). The LCN gene ortholog groups were identified according to the methods described for the water lily (*Nymphaea alba*) genome (Zhang et al., 2020). MAFFT was used to align the amino acid sequences for each gene (Katoh et al., 2005). Both concatenation and coalescent approaches were used to construct the phylogenetic trees for the dataset. RAxML was used to infer the gene trees with the coalescent approach, and then ASTRAL was used to construct species trees (Mirarab et al., 2014; Stamatakis, 2014). For each node of the species tree, the quartet score (*q*-value) was used to evaluate the reliability of these nodes. RAxML was also used to infer the ML trees from concatenated amino acid sequences for the concatenation-based analyses with ultrafast bootstrap testing (1000 replicates). The mcmctree program of PAML was used to estimate the divergence time between species/clades (with the parameters 'sample-frequency = 2, sample-number = 100 000, and burn-in = 10 000) (Yang, 2007). Based on published articles and the TimeTree website (<http://www.timetree.org>), we selected four calibration points as normal priors to restrain the age of the nodes, including 94–115 million years ago (MYA) between *O. sativa* and *A. comosus*, 148–173 MYA between *O. sativa* and *A. thaliana*, 110–124 MYA between *C. sinensis* and *A. thaliana*, and 105–115 MYA between *V. vinifera* and *A. thaliana* (Kumar et al., 2017).

Identification of whole-genome duplication events

MCScan was used to obtain the gene pairs within synteny regions from *C. praecox*, *A. trichopoda*, *L. chinense*, *C. kanehirae*, and *V. vinifera* (Tang et al., 2008; Wang et al., 2012). NCBI Blastp search with an *E*-value cutoff of 1×10^{-10} was used to obtain the best orthologous and paralogous gene pairs of transcriptions and genomes to perform the synonymous substitutions per synonymous site (K_s) analysis. The K_s values were calculated using Codeml (model = 2, runmode = -2) in the PAML package (Yang, 2007). The K_s -based phylogenetic trees were generated and WGD events

in Laurales were inferred according to the methods described for the *Litsea* genome (Chen, Li, et al., 2020).

RNA sequencing

The tepals from *C. praecox* 'Hongyun' were harvested at seven developmental stages. The middle and inner tepals for early flowering from Concolor wintersweet, Patens wintersweet, and Rubrum wintersweet were collected separately. Three biological replicates of plant materials were used for RNA sequencing. Total RNA from these tissues was extracted using RNAiso Plus (Takara, Dalian, China). Electrophoresis and an Agilent 2100 (Agilent Technologies, Santa Clara, CA, USA) were used to verify RNA quality. Finally, 39 cDNA libraries were prepared and submitted to the Illumina Novaseq 6000 platform for sequencing using paired-end cycles. The fragments per kilobase of exon model per million reads mapped (FPKM) values for genes were estimated with Hisat2 v2.0.5 and FeatureCounts v1.5.0 (Kim et al., 2019; Liao et al., 2014). The WGCNA (V1.47) package was used to perform the WGCNA (Langfelder and Horvath, 2008) and then co-expression modules were established as described by Cao, Liu, et al. (2020). Cytoscape v3.6.1 was used to visualize the co-expression networks (Kohl et al., 2011).

Sample preparation and UPLC conditions

A mixer mill (MM 400; Retsch, Haan, Germany) was used to crush the freeze-dried perianth into a fine powder with a zirconia bead for 1.5 min at 30 Hz. The powder (100 mg) was extracted overnight with 1.0 ml of 70% methanol aqueous solution at 4°C. After centrifugation at 10 000 *g* for 10 min, the extract was absorbed, filtered, and then subjected to LC-MS analysis. The UPLC-MS/MS system (MS/MS; Applied Biosystems 4500 Q TRAP, Framingham, MA, USA; UPLC, Shim-pack UFLC SHIMADZU CBM30A system) was used to analyze the sample extracts with the following analysis conditions: injection volume: 5 µl; temperature, 40°C; flow rate, 0.40 ml min⁻¹; gradient program, 95:5 v/v at 12.1 min, 5:95 v/v at 12.0 min, 5:95 v/v at 11.0 min, 95:5 v/v at 15.0 min 100:0 v/v at 0 min; solvent system, water (0.04% acetic acid):acetonitrile (0.04% acetic acid); UPLC column, Waters ACQUITY UPLC HSS T3 C18 (1.8 µm, 2.1 mm × 100 mm).

ESI-Q TRAP-MS/MS

Linear ion trap (LIT) and triple quadrupole (QQQ) scans were obtained on a QQQ-LIT mass spectrometer, API 4500 Q TRAP LC/MS/MS System (Q TRAP). The parameters of electrospray ionization (ESI) source operation were as follows: ion spray voltage, 5500 V; source temperature, 550°C; ion source, turbo spray; curtain gas, gas I (GSI), and GSI were set at 25.0, 55, and 60 psi, respectively; the collision gas was set to high. Instrument preparation tuning and mass calibration were carried out in LIT and QQQ modes using 100 and 10 µmol L⁻¹ polypropylene glycol solutions, respectively. The collision nitrogen was set to 5 psi and QQQ scans were obtained from the multiple reaction monitoring (MRM) experiments. Through further collision energy (CE) and declustering potential (DP) optimization, the CE and DP for individual MRM transitions were completed. Based on the metabolites eluted in each period, a specific set of MRM transitions were monitored for this period.

Metabolite qualification and quantification

Based on metabolite databases (i.e., HMDB, METLIN, MoToDB, MassBank, KNAPSACK, and MWDB), qualitative analysis of the primary spectrum and secondary spectrum data was conducted by MS. The quantitative analysis of metabolites was conducted from

the MRM mode of QQQ MS. R software was used to perform the unsupervised principal component analysis, hierarchical cluster analysis, and orthogonal partial least squares discriminant analysis. Metabolites with significant differences between groups were identified with the following parameters: metabolites with absolute log₂(fold change) ≥ 1 and variable importance in projection ≥ 1.

Agrobacterium-mediated transformation of tobacco

The ORFs of *CpMYB1* and *Cpmyb1*^{E201} were cloned into the pCAMBIA1304 vector using the Hieff Clone™ Plus One Step Cloning Kit (Yeasen, Shanghai, China) according to the manufacturer's instructions. The constructed recombinant vectors pCAMBIA1304-*CpMYB1* and pCAMBIA1304-*Cpmyb1*^{E201} were electroporated into *Agrobacterium tumefaciens* GV3101. The *Agrobacterium* cells were collected and then used for transformation in tobacco using the leaf disk method (Horsch et al., 1985).

Histochemical GUS assay

To determine the activation effect of *Cpmyb1*^{E201}, *CpMYB1*, and the *CpMYB1*-driven MBW complex on the *CpANS* promoter (pANS), the ORFs of *CpMYB1*, *Cpmyb1*^{E201}, *CpbHLH1*, and *CpWDR2* were cloned into the pK2GW7 vector, and the promoter sequence of *CpANS* was cloned into the pKGWFS7 vector. GUS activity in petunia petals was assayed histochemically according to the method described by Albert et al. (2011). All samples were immersed in ether for 1 min, washed twice with 50 mM phosphate buffer, subsequently washed twice with 90% acetone, and finally washed twice with 50 mM phosphate buffer, followed by incubation in GUS staining solution at 37°C overnight.

Yeast two-hybrid assay

The CDSs of *CpMYB*, *CpbHLH*, and *CpWD40* were cloned into the pGADT7-GW and pGBKT7-GW vectors, respectively, to construct the AD vectors (*CpMYB*-pGADT7 and *CpWD40*-pGADT7) and BD vectors (*CpbHLH*-pGBKT7 and *CpWD40*-pGBKT7). AD and BD plasmids were co-transformed into yeast AH109, which was spread on SD-Leu/Trp and SD-Leu/Trp/His/Ade.

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AUTHOR CONTRIBUTIONS

LZ, DY, and YC designed the study. WL, ZS, HC, and NP performed the analyses of the genome, transcriptome sequence, and metabolome. ZS, XD, MS, ZT, and CY collected the samples. ZS, YL, and ML performed the experiments. ZS, YC, and LZ wrote the manuscript. HC and NP revised the manuscript. All authors read and approved the final manuscript.

CONFLICTS OF INTEREST

The authors declare that they have no competing interests.

DATA AVAILABILITY STATEMENT

The Whole Genome Red Flower Wintersweet Project data have been deposited at GenBank under accession number JAINRJ000000000. All raw sequencing reads have been deposited in the NCBI Sequence Read Archive (<https://www.ncbi.nlm.nih.gov/sra>) under project number PRJNA704246. The datasets generated and analyzed during the current study are available from the corresponding author upon request.

SUPPORTING INFORMATION

Additional Supporting Information may be found in the online version of this article.

Figure S1. *Chimonanthus praecox* 'Hongyun' in full blossom.

Figure S2. Phylogenetic relationship between *C. praecox* and the other plant species. (a) The phylogenetic tree based on the concatenated sequence alignments of full nucleotide sequences (covering phase 0, 1, and 2 loci). (b) The phylogenetic tree based on the concatenated sequence alignments of full nucleotide sequences (covering phase 0, 1, and 2 loci) with the exclusion of *P. nigrum*. (c) The phylogenetic tree based on the concatenated sequence alignments of partial nucleotide sequences (covering phase 0 loci). (d) The phylogenetic trees based on the coalescence-based sequence alignments of full nucleotide sequences (covering phase 0, 1, and 2 loci; left) and amino acids sequences (right). The green dots with labels I, II, and III in (d) refer to phylogenetic discordances (see details in Figure S3).

Figure S3. Estimation of the 70 single-copy gene trees with different topological results based on nucleotide (a) and amino acid (b) sequences. Of the 70 trees using nucleic acid sequences (a), three topologies representing different relationship among magnoliids, monocots, and eudicots were found, i.e., the main (q1), first (q2), and second (q3) with the Q-value being 0.34, 0.33, and 0.33, respectively. For the amino acid sequence-based coalescent trees (b), three topologies representing different relationships among magnoliids, monocots, and eudicots were found, i.e., the main (q1), first (q2), and second (q3) with the Q value being 0.37, 0.35, and 0.27, respectively.

Figure S4. Summary of phylogeny and timescale of 123 plant species based on amino acid sequences of selected low-copy nuclear ortholog groups. Blue bars represent 95% credibility intervals of the estimated divergence time.

Figure S5. The coalescence-based LCN phylogenetic tree based on the nucleotide sequences of 1082 genes.

Figure S6. The coalescence-based LCN phylogenetic tree based on the nucleotide sequences of 780 genes.

Figure S7. The coalescence-based LCN phylogenetic tree based on the nucleotide sequences of 597 genes.

Figure S8. The coalescence-based LCN phylogenetic tree based on the nucleotide sequences of 99 genes.

Figure S9. The concatenated-based LCN phylogenetic tree based on the nucleotide sequences of 99 genes.

Figure S10. The coalescence-based LCN phylogenetic tree based on the amino acid sequences of 1082 proteins.

Figure S11. The coalescence-based LCN phylogenetic tree based on the amino acid sequences of 780 proteins.

Figure S12. The coalescence-based LCN phylogenetic tree based on the amino acid sequences of 597 proteins.

Figure S13. The coalescence-based LCN phylogenetic tree based on the amino acid sequences of 99 proteins.

Figure S14. The concatenated-based LCN phylogenetic tree based on 99 genes by using amino acid sequences.

Figure S15. The K_s distributions. (a–p) The K_s distributions of paralogues of *Chimonanthus salicifolius*, *Calycanthus chinensis*, *Calycanthus floridus*, *Calycanthus occidentalis*, *Idiospermum australiense*, *Cassytha filiformis*, *Cinnamomum kanehirae*, *Lindera benzoin*, *Litsea cubeba*, *Persea americana*, *Sassafras albidum*, *Peumus boldus*, *Gomortega keule*, *Piper nigrum*, *Macleaya cordata*, and *Vitis vinifera*. (q–r) The K_s distributions of orthologs between *C. praecox* and *Calycanthus filiformis*, *Lindera benzoin*, *Litsea cubeba*, *Persea americana*, *Sassafras albidum*, and *Calycanthus occidentalis*.

Figure S16. The synteny pattern between *C. praecox* and *C. salicifolius* generated by JCVI.

Figure S17. Electrophoretograms of CDSs and promoter sequences of the *CpMYB1* (*Cpr017300*) and *CpANS1* (*Cpr011668*) genes. (a) Electrophoretogram of the CDS of the *Cpr011668* gene. (b) Electrophoretogram of the CDS of the *Cpr017300* gene. (c) Electrophoretogram of the promoter sequences of the *Cpr017300* and *Cpr011668* genes.

Figure S18. Correlation analysis among differentially expressed genes, differential metabolites, and the MYB transcription factor in the phenylpropanoid pathway of different groups.

Figure S19. Overexpression of *CpMYB1* and *Cpmyb1*^{E201} in wild-type and transgenic tobacco. (a) PCR products of *CpMYB1* and *Cpmyb1*^{E201}. (b) PCR products of positive strains. (c) Positive transgenic tobacco plants. (d) GUS reporter assay of wild-type and transgenic tobacco plants.

File S1. Integrogram of quantitative evaluation of seven different anthocyanins in different samples.

Table S1. General statistics for Rubrum wintersweet based on Hi-C-assisted assembly.

Table S2. Hi-C-assisted assembly of the Rubrum wintersweet genome.

Table S3. Information statistics for Rubrum wintersweet genome assembly.

Table S4. BUSCO results for the Rubrum wintersweet genome.

Table S5. Summary of mapping statistics.

Table S6. General statistics of gene prediction.

Table S7. Statistics of repeat sequences annotated by different software.

Table S8. Statistics of repeat sequence classification.

Table S9. Statistics of non-coding RNA annotation.

Table S10. Statistical analysis of gene family clustering.

Table S11. 70 single-copy orthologous gene sets among 25 genomes.

Table S12. Low-copy nuclear (LCN) sets including 1082 genes among 123 species.

Table S13. Low-copy nuclear (LCN) sets including 780 genes among 123 species.

Table S14. Low-copy nuclear (LCN) sets including 597 genes among 123 species.

Table S15. Low-copy nuclear (LCN) sets including 99 genes among 123 species.

Table S16. K_s values of orthologous gene pairs in *C. praecox* and selected Calycanthaceae, magnoliids, and four other species.

Table S17. K_s values of orthologous gene pairs between *C. praecox* and selected Calycanthaceae, magnoliids, and four other species.

Table S18. The types and relative contents of anthocyanin metabolites in inner and middle tepals of three varieties at the early flowering stage.

Table S19. Expression of structural genes involved in the anthocyanin metabolism pathway in inner and middle tepals of three varieties at the early flowering stage.

Table S20. Cloning primers for CDSs and promoter sequences of *CpANS1* (*Cpr011668*) and *CpMYB1* (*Cpr017300*).

Table S21. CDSs and promoter sequences of *CpANS1* (*Cpr011668*) and *CpMYB1* (*Cpr017300*) in three varieties.

Table S22. 99 R2R3-MYB transcription factors in Rubrum wintersweet.

Table S23. Expression of transcription factors *Cpr017300* and *Cpr001125* in different varieties and periods.

Table S24. Co-expression of genes with *CpANS1*.

Table S25. Differentially expressed genes and differential metabolites in the phenylpropanoid pathway of five different groups between red and yellow tepals.

Table S26. PEP sequences of *CpbHLH1* (*Cpr015629*), *CpbHLH2* (*Cpr020215*), *CpWDR1* (*Cpr013636*), and *CpWDR2* (*Cpr014700*).

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